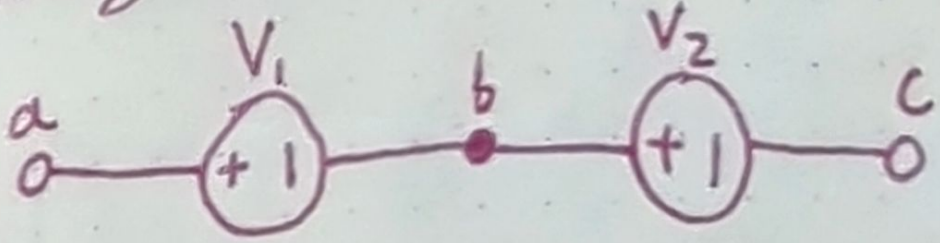
250910 Kirchhoff's Law

Adding Voltage Sources:



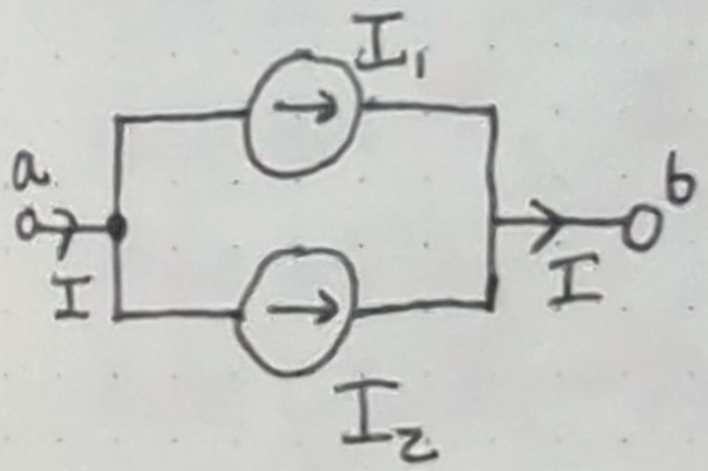
$\Rightarrow V_{ab} = V_1$

$V_{bc} = V_2$

$V_{ab} + V_{bc} = V_{ac} \Rightarrow V_{ac} = V_1 + V_2$

Voltage Sources add in Series

Adding Current Sources:



Look at node a:

$I = I_1 + I_2$

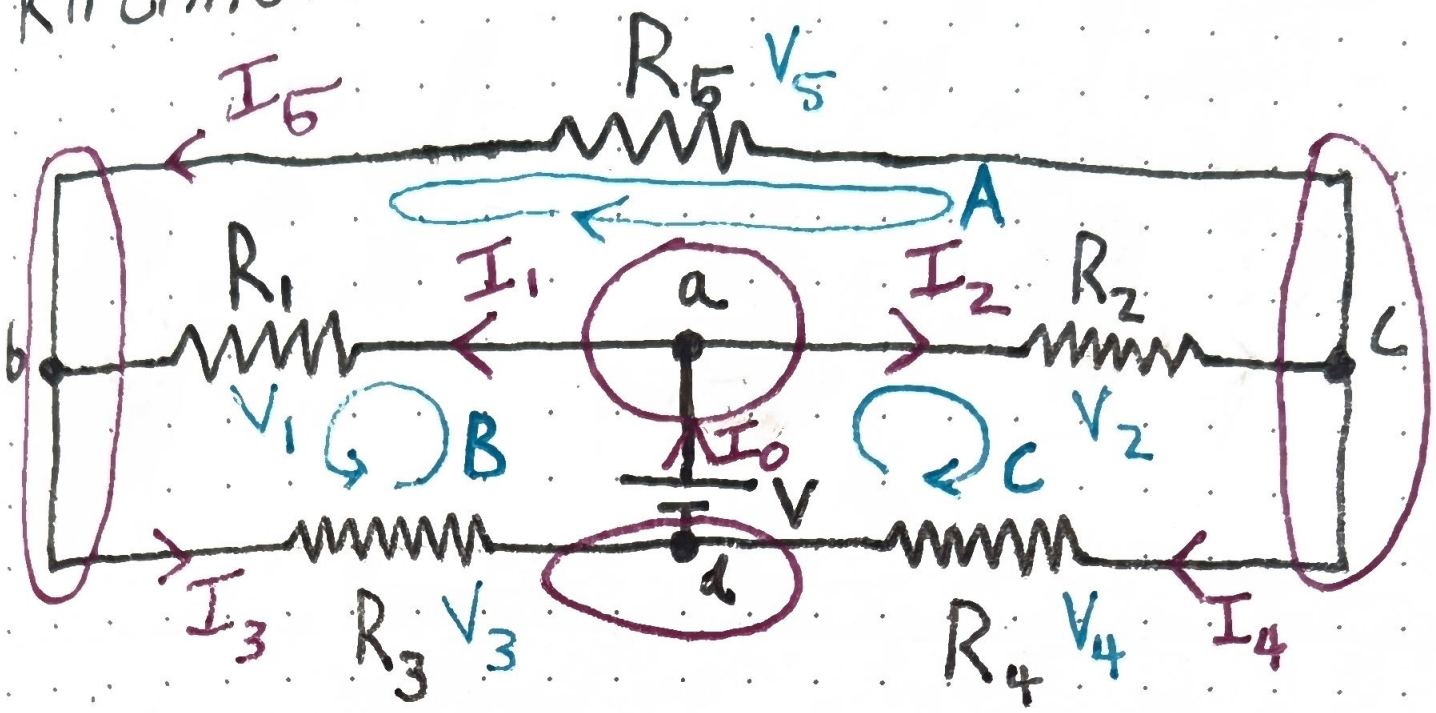
Look at node b:

$I = I_1 + I_2$

The same

Current Sources add in Parallel

Kirchhoff:



Kirchhoff Current Law (KCL):

Node a: $I_0 - I_1 - I_2 = 0$

Node b: $I_1 - I_3 + I_5 = 0$

Node c: $I_2 - I_4 - I_5 = 0$

Node d: $-I_0 + I_3 + I_4 = 0$

Kirchhoff Voltage Law (KVL):

Loop A: $-V_1 + V_5 + V_2 = 0$

Loop B: $-V_1 - V_3 + V = 0$

Loop C: $-V_2 - V_4 + V = 0$